

Vikash Kumar

Adm. No. U24CS145 — 9942921085 — vikashonly88@gmail.com

[linkedin.com/in/vikash-kumar-b50a5032b](https://www.linkedin.com/in/vikash-kumar-b50a5032b) — github.com/Vikash88Kumar



Education

Sardar Vallabhbhai National Institute of Technology, Surat

Surat, Gujarat

Bachelor of Technology in Computer Science and Engineering **CGPA: 7.71**

June 2028 (Expected)

Relevant Coursework: Intro to C/C++, DSA, Discrete Mathematics, OOPs, DBMS, Computer Networks.

Competitive Programming:

- **LeetCode:** leetcode.com/u/vikashkr05 — Solved 300+ Data Structures & Algorithms problems.
- **Codeforces:** codeforces.com/profile/vikashkr05 — Current Rank: Newbie (Max Rating: 1000+).

Projects

HALO — AI Sports Video Analytics & Multi-Object Tracking Engine

[GitHub](#) — Python, PyTorch,

YOLOv11, ByteTrack, OpenCV, Scikit-Learn

- Developed an end-to-end sports video analytics platform tracking 22+ entities concurrently in high-definition match footage using fine-tuned YOLO and ByteTrack.
- Built a STN-ResNet18 digit OCR model with temporal voting to recognize 2-digit jersey numbers on cropped player torso images with over 91% accuracy.
- Created a homographic pitch-projection pipeline transforming broadcast camera views into dynamic 2D tactical radars to compute real-time spatial positioning, velocity, and team analytics.
- Optimized inference and video encoding via multi-threaded OpenCV & FFmpeg H.264 pipelines, accelerating video rendering throughput by 3.5x.

Notification Engine SaaS

[GitHub](#) — FastAPI, Kafka, PostgreSQL, Redis, Docker, Firebase, Resend — Present

- Architected an event-driven notification microservice using FastAPI, Apache Kafka, and Redis, capable of asynchronously routing and delivering 10,000+ notifications/sec across Email, FCM Push, and Real-Time WebSockets.
- Designed a multi-channel fan-out architecture consuming from a unified notification.events topic and routing payload streams to dedicated Kafka worker queues (email.queue, push.queue, inapp.queue) with j50ms orchestration latency.
- Engineered a Smart Presence Suppression Engine utilizing Redis TTL caching and WebSocket heartbeat tracking to suppress redundant Push/Email spam when users are online, reducing unnecessary 3rd-party API calls by 35%.
- Implemented an enterprise fault-tolerance framework featuring an Exponential Backoff Retry Worker (3 retry attempts) and a Dead Letter Queue (DLQ) topic for failed payload isolation, achieving 99.9% delivery reliability.
- Scaled real-time in-app delivery across multi-node server instances by establishing a Redis Pub/Sub message broker, eliminating single-instance WebSocket bottlenecks.

Smart Crop Recommendation System

[GitHub](#) — Scikit-learn, Flask, REST API, Docker, React (Vite) — Nov

2025

- Developed an end-to-end machine learning pipeline using pandas, scikit-learn, and XGBoost to predict optimal crops from soil parameters, incorporating feature engineering and cross-validation for robust predictions, **achieving 94% accuracy across 10,000+ records**.
- Deployed the trained model as a Dockerized Flask REST API and integrated it with a React (Vite) frontend via Axios to deliver real-time, location-aware crop recommendations, **maintaining under 200ms API response time**.
- Automated data preprocessing, validation, and reproducible training workflows, improving model reliability, scalability, and data-driven decision support for agricultural applications.

Multi Social Media Web Application

[GitHub](#) — React, Node.js, Express, MongoDB, Socket.IO, JWT — Present

- Engineered a full-stack platform featuring video streaming, micro-blogging, and real-time chat, scaling to support user interaction across multiple formats **for up to 500+ concurrent connections**.
- Developed a scalable Node.js and Express backend exposing **20+ secure REST APIs** for robust JWT authentication, media uploads, and dynamic user profiles.
- Built a real-time messaging engine via Socket.IO, achieving **sub-50ms latency** to support private chat rooms, persistent history, and live read receipts without page refreshes.
- Optimized MongoDB aggregation pipelines for customized user feeds and comprehensive engagement metrics, effectively **reducing database query execution time by nearly 40%**.

Technical Skills

Programming & Markup Languages: C, C++, JavaScript, Python, SQL

Frameworks & Libraries: React.js, Redux Toolkit, Tailwind CSS, Node.js, Express.js, FastAPI, MongoDB, Mongoose, Socket.IO

Software & Tools: Git/GitHub, Docker, Redis, Apache Kafka, Firebase, Resend, Postman, Hugging Face

Achievements

- **Winners – Computer Vision Track:** Code Cup 2026 — ACM RVCE.
- Secured 23rd Rank in GDGC Kaggle Datathon organised by GDGC NIT Surat.
- Secured 3rd position in CAN-SAT.
- Selected as a Semi-Finalist at NEXUS-WEB WONDERS.

Social Engagements

Executive: NEXUS, GDGC, Chintan Pratisthan.